

|          | AMPACITY | MVA RATING | COND TEMP | SHEATH TEMP | TOTAL LOSS | STATUS    |
|----------|----------|------------|-----------|-------------|------------|-----------|
| BURIED   | 822 A    | 187.88 MVA | 90.0 °C   | 78.7 °C     | 34.919 W/m | Converged |
| FREE AIR | 991 A    | 226.56 MVA | 90.0 °C   | 73.6 °C     | 50.770 W/m | Converged |

1. Design Data & Input Parameters

| Parameter                         | Formula / Symbol | Value                  | Unit  |
|-----------------------------------|------------------|------------------------|-------|
| Conductor material                |                  | Cu                     |       |
| Conductor size                    |                  | 630                    | mm2   |
| Conductor type                    |                  | stranded_compact       |       |
| Conductor diameter Dc             |                  | 30.3                   | mm    |
| Conductor screen OD Dc_sc         |                  | 33.3                   | mm    |
| Max conductor temp theta_max      |                  | 90                     | degC  |
| Insulation material               |                  | XLPE                   |       |
| Insulation outer diameter Di      |                  | 64.3                   | mm    |
| Insulation screen OD Di_sc        |                  | 66.9                   | mm    |
| Insulation thickness t_ins        |                  | 15.50                  | mm    |
| Relative permittivity er          |                  | 2.5000                 |       |
| Loss tangent tan(delta)           |                  | 0.001000               |       |
| Screen type                       |                  | Al                     |       |
| Screen mean diameter ds           |                  | 67.7                   | mm    |
| Screen area As                    |                  | 170.2                  | mm2   |
| Armour type                       |                  | none                   |       |
| Oversheath material               |                  | PE                     |       |
| Cable outer diameter De           |                  | 75.5                   | mm    |
| Rated voltage U                   |                  | 132.0                  | kV    |
| U0 line-to-earth                  |                  | 76.210                 | kV    |
| Frequency                         |                  | 50                     | Hz    |
| Bonding                           |                  | solid                  |       |
| Formation                         |                  | trefoil                |       |
| Depth to cable centre L           |                  | 1000                   | mm    |
| Axial spacing s                   |                  | 75.5                   | mm    |
| Ambient ground temp theta_a       |                  | 20                     | degC  |
| Soil thermal resistivity rho_soil |                  | 1.00                   | K.m/W |
| Ambient air temp theta_a_air      |                  | 25                     | degC  |
| Air mounting                      |                  | free                   |       |
| Solar radiation                   |                  | H=1000 W/m²,<br>σ=0.40 |       |
| Wind speed                        |                  | None                   |       |

2. DC Resistance [IEC 60287-1-1 Cl.2.1.1]

| Parameter                   | Formula / Symbol | Value      | Unit  |
|-----------------------------|------------------|------------|-------|
| IEC 60228 R0 at 20 degC     |                  | 2.8300e-05 | ohm/m |
| Temperature coefficient a20 |                  | 3.9300e-03 | 1/K   |
| R0 at theta_amb = 20 degC   |                  | 2.8300e-05 | ohm/m |
| R0 at theta_max = 90 degC   |                  | 3.6085e-05 | ohm/m |

### 3. AC Resistance at theta\_max [IEC 60287-1-1 Cl.2.1.2, 2.1.4]

| Parameter                                     | Formula / Symbol | Value            | Unit  |
|-----------------------------------------------|------------------|------------------|-------|
| ks strand constant (IEC 60287-1-1 Table 2)    |                  | 1.0000           |       |
| kp proximity constant (IEC 60287-1-1 Table 2) |                  | 1.0000           |       |
| $xs^2 = ks \cdot (8\pi f / R0) \cdot 1e-7$    |                  | 3.4824042386     |       |
| $xs_{\text{eff}} = (xs^2)^2$                  |                  | 12.1271392811    |       |
| ys skin effect factor                         |                  | 0.0601241268     |       |
| $xp_{\text{eff}} = (xp^2)^2$                  |                  | 12.1271392811    |       |
| yp" intermediate proximity (s=spacing_s)      |                  | 0.0601241268     |       |
| yp proximity factor (s=spacing_s)             |                  | 0.0351000648     |       |
| R at theta_max AC (buried/air)                |                  | 3.9521526380e-05 | ohm/m |
| yp proximity factor (s=Do duct)               |                  | 0.0101077563     |       |
| R at theta_max AC (duct)                      |                  | 3.8619670681e-05 | ohm/m |

### 4. Dielectric Losses [IEC 60287-1-1 Cl.2.2]

| Parameter                                                | Formula / Symbol | Value            | Unit  |
|----------------------------------------------------------|------------------|------------------|-------|
| Relative permittivity er                                 |                  | 2.5000           |       |
| Loss tangent tan(delta)                                  |                  | 0.001000         |       |
| U0 line-to-earth                                         |                  | 76.2100          | kV    |
| Di insulation outer diameter                             |                  | 64.3000          | mm    |
| Dc_sc conductor screen OD                                |                  | 33.3000          | mm    |
| Capacitance C = $er / (18 \ln(Di/Dc\_sc)) \cdot 10^{-9}$ |                  | 0.211077         | nF/m  |
| omega = 2 pi f                                           |                  | 314.159265       | rad/s |
| Wd = omega C U0^2 tan(delta)                             |                  | 3.8513583657e-01 | W/m   |
| TB880 GP7                                                |                  | Applied          |       |

### 5. Screen / Sheath Setup [IEC 60287-1-1 Cl.2.3]

| Parameter                       | Formula / Symbol | Value      | Unit  |
|---------------------------------|------------------|------------|-------|
| Screen material                 |                  | Al         |       |
| Screen resistivity rho_s        |                  | 2.8400e-08 | ohm.m |
| Screen area As                  |                  | 170.1500   | mm2   |
| Wire/sheath outer OD D_sc_outer |                  | 68.5000    | mm    |
| Lay factor LFs                  |                  | 1.000000   |       |
| Screen Rs0 at 20 degC           |                  | 1.6691e-04 | ohm/m |
| X_sheath (s=spacing_s buried)   |                  | 5.0403e-05 | ohm/m |
| X_sheath_duct (s=Do duct)       |                  | 8.9203e-05 | ohm/m |

### 6. Armour Losses [IEC 60287-1-1 Cl.2.4]

| Parameter   | Formula / Symbol | Value    | Unit |
|-------------|------------------|----------|------|
| Armour type |                  | None     |      |
| lam2        |                  | 0.000000 |      |

## 7. Thermal Resistances [IEC 60287-2-1]

| Parameter                                      | Formula / Symbol | Value           | Unit  |
|------------------------------------------------|------------------|-----------------|-------|
| rho_ins insulation                             |                  | 3.50            | K.m/W |
| rho_sc semiconducting screens                  |                  | 2.50            | K.m/W |
| rho_tape bedding tape under wire screen        |                  | 6.00            | K.m/W |
| T1_cs conductor screen                         |                  | 0.0375644199    | K.m/W |
| T1_i insulation                                |                  | 0.3665350785    | K.m/W |
| T1_si insulation screen                        |                  | 0.0157719906    | K.m/W |
| T1_uwbt bedding tape under wire screen         |                  | 0.0000000000    | K.m/W |
| T1_raw = sum of all layers                     |                  | 0.4198714890    | K.m/W |
| T1 correction x1.00 (foil=False, cover=100.0%) |                  | 0.4198714890    | K.m/W |
| T2 buried (screen to foil tape)                |                  | 0.0000000000    | K.m/W |
| T3 reference diameter D_start                  |                  | 68.5000         | mm    |
| T3_raw oversheath (single-layer)               |                  | 0.0541996092    | K.m/W |
| T3 buried x1.6 (touching trefoil factor)       |                  | 0.0867193748    | K.m/W |
| T3 air x1.0 (no touching factor in air)        |                  | 0.0541996092    | K.m/W |
| u = 2L/De (buried)                             |                  | 26.490066       |       |
| T4 buried (soil)                               |                  | 1.5946928925    | K.m/W |
| T4 air (iterative — see air section)           |                  | see air section |       |

## 8. Electrical Parameters

| Parameter             | Formula / Symbol | Value    | Unit   |
|-----------------------|------------------|----------|--------|
| Inductance L          |                  | 0.160439 | uH/m   |
| Reactance X = omega L |                  | 50.4033  | uOhm/m |
| Pos-seq resistance R1 |                  | 51.1372  | uOhm/m |
| Pos-seq impedance Z1  |                  | 71.8018  | uOhm/m |
| Capacitance C         |                  | 0.211077 | nF/m   |
| Charging current Ic   |                  | 5.0536   | mA/m   |

## 9. Sheath Losses at Convergence (Buried)

| Parameter              | Formula / Symbol | Value        | Unit  |
|------------------------|------------------|--------------|-------|
| theta_s at convergence |                  | 78.712987    | degC  |
| Rs at theta_s          |                  | 2.0641e-04   | ohm/m |
| X sheath reactance     |                  | 5.0403e-05   | ohm/m |
| lam1' circulating      |                  | 0.2939065057 |       |
| lam1" eddy             |                  | 0.0000000000 |       |
| lam1 = lam1' + lam1"   |                  | 0.2939065057 |       |
| lam2 armour            |                  | 0.000000     |       |

## 10. Ampacity — Direct Buried [IEC 60287-1-1 Cl.1.4.1.1]

| Parameter                 | Formula / Symbol | Value          | Unit |
|---------------------------|------------------|----------------|------|
| Delta_theta               |                  | 70.00          | K    |
| Etop numerator            |                  | 69.2715741012  |      |
| Ebot denominator          |                  | 1.0258e-04     |      |
| I = sqrt(Etop/Ebot) exact |                  | 821.7758159100 | A    |
| I rounded TB880 GP2       |                  | 822            | A    |
| MVA = sqrt(3) x U x I     |                  | 187.8832       | MVA  |
| Iterations                |                  | 5              |      |
| Convergence               |                  | YES            |      |

### 11. Temperature Profile — Buried

| Parameter                | Formula / Symbol | Value     | Unit |
|--------------------------|------------------|-----------|------|
| theta_c conductor target |                  | 90.0      | degC |
| theta_c back-calculated  |                  | 90.000000 | degC |
| theta_s sheath surface   |                  | 78.712987 | degC |
| theta_j cable surface    |                  | 75.684845 | degC |
| theta_a ambient          |                  | 20.0      | degC |

### 12. Loss Budget — Buried

| Parameter     | Formula / Symbol | Value        | Unit |
|---------------|------------------|--------------|------|
| Conductor Wc  |                  | 26.689499    | W/m  |
| Sheath Ws     |                  | 7.844217     | W/m  |
| Armour Wa     |                  | 0.000000     | W/m  |
| Dielectric Wd |                  | 3.851358e-01 | W/m  |
| Total         |                  | 34.918852    | W/m  |

### 13. Free Air Parameters [IEC 60287-2-1 Cl.2.2.1]

| Parameter            | Formula / Symbol | Value        | Unit                                 |
|----------------------|------------------|--------------|--------------------------------------|
| Formation            |                  | trefoil      |                                      |
| Mounting             |                  | free         |                                      |
| Z (IEC Table 2)      |                  | 0.9600       |                                      |
| E (IEC Table 2)      |                  | 1.2500       |                                      |
| g (IEC Table 2)      |                  | 0.2000       |                                      |
| De* (m)              |                  | 0.075500     | m                                    |
| $h = Z/(De^*)^g + E$ |                  | 2.8594661337 | W/(m <sup>2</sup> K <sup>5/4</sup> ) |
| Solar                |                  | Yes          |                                      |
| H                    |                  | 1000.0       | W/m <sup>2</sup>                     |
| sigma                |                  | 0.40         |                                      |
| theta_a_air          |                  | 25.0         | degC                                 |
| Delta_theta          |                  | 65.00        | K                                    |

### 14. Free Air T4 [IEC 60287-2-1 Cl.2.2.1]

| Parameter                                 | Formula / Symbol | Value         | Unit               |
|-------------------------------------------|------------------|---------------|--------------------|
| $x = (\theta_j - \theta_a)^{1/4}$         |                  | 2.6024876298  | K <sup>(1/4)</sup> |
| Dtheta_s = theta_j - theta_a              |                  | 45.8727414814 | K                  |
| $T4 = 1/(\pi \cdot De^* \cdot h \cdot x)$ |                  | 0.5665386515  | K.m/W              |
| Solar sigma*De*H*T4                       |                  | 17.1094672748 | K                  |

15. Sheath Losses at Convergence (Air)

| Parameter              | Formula / Symbol | Value        | Unit  |
|------------------------|------------------|--------------|-------|
| theta_s at convergence |                  | 73.624466    | degC  |
| Rs at theta_s          |                  | 2.0298e-04   | ohm/m |
| X sheath reactance     |                  | 5.0403e-05   | ohm/m |
| lam1'                  |                  | 0.2982916317 |       |
| lam1"                  |                  | 0.0000000000 |       |
| lam1                   |                  | 0.2982916317 |       |
| lam2                   |                  | 0.000000     |       |

16. Ampacity — Free Air [IEC 60287-1-1 Cl.1.4.4.1]

| Parameter                   | Formula / Symbol | Value          | Unit |
|-----------------------------|------------------|----------------|------|
| Etop numerator (incl solar) |                  | 47.5706103973  |      |
| Ebot denominator            |                  | 4.8444e-05     |      |
| I = sqrt(Etop/Ebot) exact   |                  | 990.9411775310 | A    |
| I rounded TB880 GP2         |                  | 991            | A    |
| MVA = sqrt(3) x U x I       |                  | 226.5596       | MVA  |
| Iterations                  |                  | 5              |      |
| Convergence                 |                  | YES            |      |

17. Temperature Profile — Free Air

| Parameter                | Formula / Symbol | Value     | Unit |
|--------------------------|------------------|-----------|------|
| theta_c conductor target |                  | 90.0      | degC |
| theta_c back-calculated  |                  | 90.000000 | degC |
| theta_s sheath surface   |                  | 73.624466 | degC |
| theta_j cable surface    |                  | 70.872741 | degC |
| theta_a ambient air      |                  | 25.0      | degC |

18. Loss Budget — Free Air

| Parameter     | Formula / Symbol | Value        | Unit |
|---------------|------------------|--------------|------|
| Conductor Wc  |                  | 38.808733    | W/m  |
| Sheath Ws     |                  | 11.576320    | W/m  |
| Armour Wa     |                  | 0.000000     | W/m  |
| Dielectric Wd |                  | 3.851358e-01 | W/m  |
| Solar heat    |                  | 17.109467    | W/m  |
| Total         |                  | 50.770189    | W/m  |

19. Electrical Parameters

| Parameter             | Formula / Symbol | Value    | Unit   |
|-----------------------|------------------|----------|--------|
| Inductance L          |                  | 0.160439 | uH/m   |
| Reactance X = omega L |                  | 50.4033  | uOhm/m |
| Pos-seq resistance R1 |                  | 51.1372  | uOhm/m |
| Pos-seq impedance Z1  |                  | 71.8018  | uOhm/m |
| Capacitance C         |                  | 0.211077 | nF/m   |
| Charging current Ic   |                  | 5.0536   | mA/m   |

GP2: ampacity rounded down. GP6: eddy losses always calculated. GP7: dielectric losses always calculated. GP8: exact T4 formula. T3: 1.6x factor for touching trefoil (buried only).  
Solar:  $H=1000 \text{ W/m}^2$ ,  $\sigma=0.40$ .  
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